CORE PURE / FURTHER CALCULUS

# Integration by Parts

Questions by topic 15 questions · 105 marks

## Questions


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|---|---|---|
| Question 1 | *5 marks* | 2 |
| Question 2 | *6 marks* | 3 |
| Question 3 | *7 marks* | 4 |
| Question 4 | *5 marks* | 5 |
| Question 5 | *7 marks* | 6 |
| Question 6 | *7 marks* | 7 |
| Question 7 | *7 marks* | 8 |
| Question 8 | *9 marks* | 9 |
| Question 9 | *9 marks* | 10 |
| Question 10 | *6 marks* | 11 |
| Question 11 | *6 marks* | 12 |
| Question 12 | *6 marks* | 13 |
| Question 13 | *6 marks* | 14 |
| Question 14 | *7 marks* | 15 |
| Question 15 | *12 marks* | 16 |

**1.** Use integration by parts to find

$$\int_1^9 \sqrt{x} \ln\left(\frac{9}{x}\right) \mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where $a$ and $b$ are constants to be found. **[5]**

**2.** Given that

$$y = \arctan\left(\frac{x-2}{x+2}\right)$$

where $x \neq -2$

**(a)** find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

**(b)** Use integration by parts to find

$$\int \arctan\left(\frac{x-2}{x+2}\right) \mathrm{d}x$$ **[4]**

**3.** Use integration by parts to find

$$\int_0^2 x^2 \ln(1+x)\,\mathrm{d}x$$

giving your answer in the form $a + b\ln 3$, where $a$ and $b$ are constants to be found. **[7]**

**4.** Use integration by parts to find

$$\int \operatorname{arsinh}(2x+1)\,\mathrm{d}x$$

giving your answer in terms of logarithms. **[5]**

**5.**

$$h(\theta) = (3\sin\theta - \cos\theta)(\sin\theta + 2\cos\theta)$$

**(a)** Show that $h(\theta) = \dfrac{1}{2} - \dfrac{5}{2}\cos 2\theta + \dfrac{5}{2}\sin 2\theta$. **[2]**

**(b)** Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \theta\right) h(\theta)\,\mathrm{d}\theta$$ **[5]**

**6.** **(a)** Find

$$\int x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[5]}$$

**(b)** Hence find the exact value of

$$\int_0^1 x^2 \arctan x \, \mathrm{d}x \qquad \textbf{[2]}$$

**7.** **(a)** Find

$$\int x^3 \arctan x \, \mathrm{d}x$$ **[5]**

**(b)** Hence find the exact value of

$$\int_0^{\sqrt{3}} x^3 \arctan x \, \mathrm{d}x$$ **[2]**

**8.** **(a)** Using the logarithmic form of $\operatorname{artanh} x$, prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

for $|x| < 1$. **[3]**

**(b)** Hence find

$$\int_0^{1/2} \operatorname{artanh} x \, \mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

**(c)** Priya tries to evaluate

$$\int_0^{3/2} \operatorname{artanh} x \, \mathrm{d}x$$

using her calculator, and gets an 'error'. Explain why. **[1]**

**9.** **(a)** Using the logarithmic form

$$\operatorname{artanh} x = \frac{1}{2} \ln \left( \frac{1+x}{1-x} \right), \qquad |x| < 1$$

prove that the derivative of $\operatorname{artanh} x$ is

$$\frac{1}{1-x^2}$$

**[3]**

**(b)** Hence find

$$\int_0^{1/3} \operatorname{artanh}(2x) \, \mathrm{d}x$$

giving your answer in exact logarithmic form. **[5]**

**(c)** Noah tries to evaluate

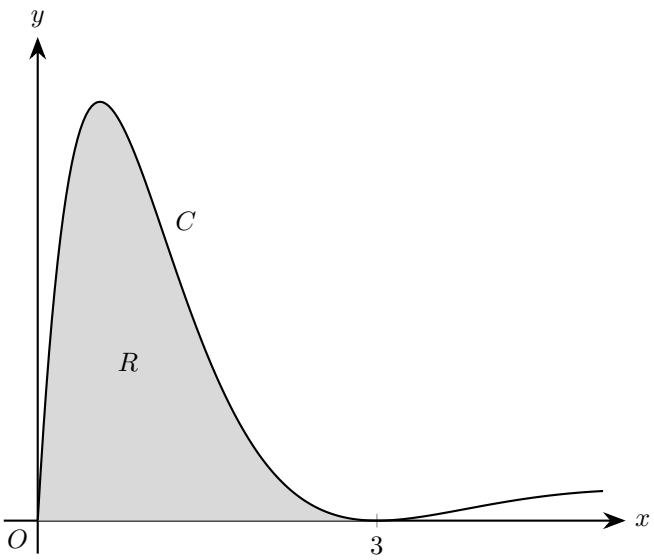
$$\int_0^{1/2} \operatorname{artanh}(2x) \, \mathrm{d}x$$

using his calculator, and gets an error. Explain why. **[1]**

**10.** For $0 < x < 1$, use integration by parts to find

$$\int x \operatorname{artanh}(2x-1)\,\mathrm{d}x$$

giving your final answer in terms of logarithms. **[6]**




**11.** The sketch shows part of the curve $C$ given by

$$y = x(3-x)^2e^{-x} \qquad x \geq 0$$

The shaded region $R$ is bounded by $C$ and the $x$-axis for $0 \leq x \leq 3$.

Find the exact area of $R$, giving your answer in the form

$$A + Be^{-3}$$

where $A$ and $B$ are rational numbers. **[6]**

**12.** Show that

$$\int_{e^{-1/2}}^{1} x(\ln x)^2 \, \mathrm{d}x = a + \frac{b}{e}$$

where $a$ and $b$ are rational constants to be found. **[6]**

**13.** Given that

$$y = \arcsin\left(\frac{x-1}{3}\right)$$

**(a)** find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in its simplest form. **[2]**

**(b)** Use integration by parts to find

$$\int \arcsin\left(\frac{x-1}{3}\right)\,\mathrm{d}x$$ **[4]**

**14.**

$$g(\theta) = (1 + \sin\theta - \cos\theta)^2$$

**(a)** Show that $g(\theta) = 2 + 2\sin\theta - 2\cos\theta - \sin 2\theta$. **[2]**

**(b)** Hence, using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \theta g(\theta)\,\mathrm{d}\theta$$ **[5]**

**15.** **(i)** Sketch the graph of $y = \arctan x$ for $x \in \mathbb{R}$, indicating any horizontal asymptotes. **[1]**

**(ii)** Prove that

$$\frac{\mathrm{d}}{\mathrm{d}x}(\arctan x) = \frac{1}{1+x^2} \quad \textbf{[3]}$$

**(iii)** Using integration by parts and suitable substitutions, show that

$$\int_0^1 x(\arctan x)^2 \,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2 \quad \textbf{[8]}$$